Demo: Kelvin Water Dropper

A no-spark test of electrostatic feedback

Adult-built and adult-operated — measure the rise, never wait for a spark

Lesson Demo B · verification demonstration · after Lesson 1 — gravitational input only

THIS IS NOT A LEARNER CONSTRUCTION

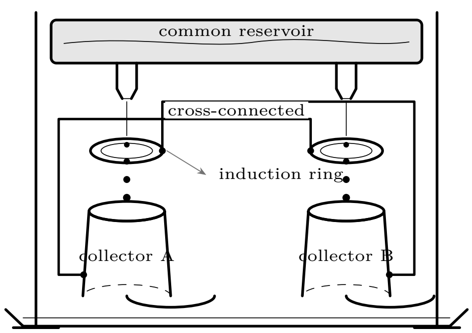
A Kelvin dropper can accumulate high electrostatic voltage without a battery. The learner predicts, watches from the boundary, and records remote readings. An adult builds and operates the apparatus. Stop the water before approaching; use only the installed high-resistance discharge path; verify zero before touching. A snap, spark, hair movement, or sensation is a failed run, not a result.

What you need

- Adult-built two-stream dropper on a rigid insulating frame
- Two induction rings and two deep metal collectors
- Matched plastic nozzles and a common reservoir
- Spill tray wider than the entire apparatus
- Electroscope with a fixed millimetre scale, readable at a distance
- Stopwatch, two measuring cups, ruler, towels, and run sheet
- Permanently arranged high-resistance discharge lead

Predict first

Choose your evidence before the water runs. Would charge separation be better shown by one sudden leaf movement, by a steadily increasing deflection, or by a spark? Will the apparatus always choose the same polarity? What must be measured to rule out "one side simply received more water"?



A collector biases the opposite ring.
All water and all conductors remain inside the tray.

Match Equalize water head, flows, ring positions, and dry supports. Otherwise geometry can mimic charge.

Breakup is critical Each jet must separate into drops while it is inside its ring. A continuous water path carries charge back toward the reservoir.

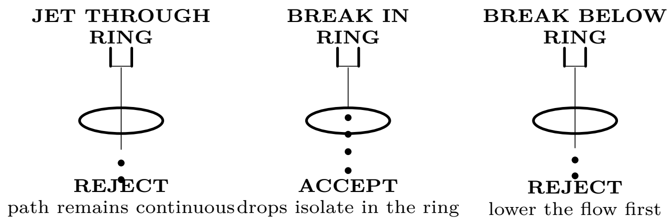
No assigned signs A minute initial imbalance selects the polarities. The labels A and B identify sides, not positive and negative terminals.

The output is a trace Electroscope deflection versus time is the evidence. Spark distance is neither required nor permitted.

Set the ceiling before the run

The adult writes the electroscope stop mark and the 20-second time limit on the run sheet before removing the discharge lead. The first limit reached ends the run. Neither limit may be raised because "it is nearly there." Also write the observer's position and confirm the scale is legible from it; a reading that requires approaching the apparatus is not an admissible reading.

Three geometry states



Adult preflight: water off

1. Mark the observation boundary. Confirm the learner can read the electroscope and stopwatch without crossing it.
2. Push gently on the frame, reservoir, rings, and collectors. Nothing may rock, slide, or sag. Confirm the tray contains every plausible drip.
3. Trace the connections with a finger held clear: collector A goes to ring B; collector B goes to ring A. No conductor may approach the other side.
4. Inspect the dry insulators and dry surroundings. Confirm there is no nearby flame, solvent vapour, charged apparatus, open mains equipment, or thunderstorm activity.
5. Demonstrate that the electroscope rests at zero and responds to its safe check source. Set a conservative stop line well below its scale limit.
6. Inspect the fixed high-resistance discharge path. State the stop order aloud: **water off; last drop landed; discharge; verify zero; approach.**

First prove the water is matched

Keep charge from accumulating using the apparatus's designed discharged state. Run both streams for exactly 10 seconds into separate measuring cups. Count drops and measure volume. Repeat once. Adjust flow only with water stopped. Both break points must remain inside their rings and every drop must enter its collector.

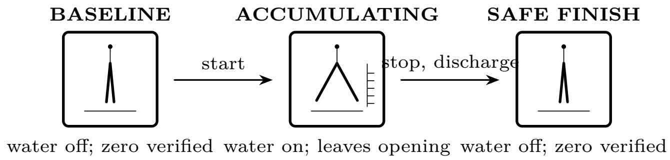
Ask before continuing: Are the two sides close enough that a deflection cannot reasonably be blamed on grossly unequal delivery? Did either break point move during the count? What number, rather than an impression, supports the answer? Record both control runs on the worksheet at the end.

Measure the feedback

With water off, discharge through the installed resistor and confirm zero. Park the lead. Start both streams and the stopwatch together. From behind the boundary, record leaf separation in millimetres every five seconds. End the run at 20 seconds or at the stop line, whichever occurs first. Stop and discharge in the stated order. Repeat from zero; never "continue" the first run. Use the end worksheet for both traces and any early-stop time.

Read the evidence

A credible result has a zero baseline, a gradual rise during both isolated runs, and a return to zero after each controlled discharge. One twitch is not a trend. A large final reading without intermediate measurements is not a feedback trace.



Questions for the record. Did both traces rise monotonically? Did one begin later? Could wet insulation or an off-centre stream explain that difference? Why does the return to zero belong in the evidence rather than only in the safety procedure? Where did the electrical energy come from?

One controlled change

After a complete zero verification, reduce *both* drip rates while keeping both breakup points within their rings. Repeat the same 20-second measurement. Compare the change in millimetres per five seconds. Do not compare maximum voltage and do not add, narrow, or test a spark gap.

Troubleshoot only in the safe state

Observation	After water off, discharge, and verified zero
No repeatable movement	Check the electroscope with its safe test; trace both cross-connections; dry the tray and every insulator.
Only one stream forms drops	Match the flow and reposition its breakup point inside the ring. Never move a live ring.
Drops touch a ring	Re-centre the nozzle and ring; push-test the frame again.
Fast jump toward stop line	End the run, increase conductor separation, and choose a lower stop mark before deciding whether to repeat.
First run works, second is flat	Restore the measured water control; look for wet supports, residual charge, and a shifted stream.
Any snap, spark, or sensation	Demonstration failed. Discharge, document the event, and do not run again that session.

Final proof — all conditions required

Mark the demonstration **verified** only when two runs start at indicated zero, show several increasing millimetre readings before the conservative stop line, keep both breakup points inside their rings, lose no water outside the collectors, produce no spark or snap, and finish at indicated zero after the installed high-resistance discharge. If any condition is missing, record **not verified**. A spectacular event never repairs missing evidence.

Run worksheet

Apparatus ID: ______________ **Date:** ______________ **Adult operator:** ______________

Predicted evidence. Before the run, state what visible, measurable pattern would support electrostatic feedback without a spark.

Limits fixed before operation

Electroscope stop mark	______	Time limit	______
Observer position	______	Scale legible there?	yes / no

Hydraulic control — discharged state

Control run	A drops/10 s	B drops/10 s	A mL/10 s	B mL/10 s
1	______	______	______	______
2	______	______	______	______

Breakup stayed inside both rings: **yes / no** All drops entered collectors: **yes / no**

Is the flow match adequate to continue? **yes / no**. Cite the numbers, not appearance:

Electroscope trace — isolated state

Run	0 s	5 s	10 s	15 s	20 s	stop reason/time
1, mm	______	______	______	______	______	______
2, mm	______	______	______	______	______	______
slower drip, mm	______	______	______	______	______	______

Evidence record

Did both standard traces rise through several readings? **yes / no** Did each finish at verified zero? **yes / no**

Strongest alternative explanation still unresolved:

Claim supported by the measurements:

Final disposition: **verified / not verified / stopped on safety condition**